Stat 175 Final Project Report

Debiasing Estimates for RDS in Practice

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Abstract

Respondent-Driven Sampling (RDS) is a useful statistical sampling tool for gathering data from target populations that are difficult to otherwise study due to sparsity, stigma, and small population sizes. However, the application of RDS in practice involves strong assumptions, specifically the assumption that homophilic communities within the target population will eventually be sufficiently mixed in the sampling chain to reduce bias induced by uneven seed selection and preferential recruitment. This paper tests that assumption in practice through a study of anti-AI Harvard students, utilizing the standard Volz-Heckathorn estimator as well as a novel estimator to reduce RDS bias in practice. Our proposed α-weighted correction provides a principled way to account for higher level dependencies in sampling, and gives modest improvements in simulation. An application of this methodology on the anti-AI dataset yields moderate reduction in cross-group heterogeneity, suggesting that this method could improve the recovery of true population-level distributions.

Data and methodology in code is included in this GitHub repository.

1. INTRODUCTION AND MOTIVATION

Surveys of "hidden" or "hard-to-reach" populations face several structural obstacles. These populations often face stigma, marginalization, or are otherwise rare, and they lack manageable sampling frames with known size and boundaries. Their members are too sparsely distributed in the general population to be reached efficiently through conventional survey design, and privacy concerns may result in refusal to participate or unreliable answers. Drug injectors in Connecticut constitute one such population. Introduced by Heckathorn [1] to study this population's characteristics, Respondent-Driven Sampling (RDS) and its extensions by Salganik and Heckathorn [3] , Heckathorn [2] [4] , and Volz and Heckathorn [5] , address these obstacles, as Gile and Handcock [6] frame, through innovating in two areas: **survey design** and **inference**.

1.1. Survey Design

The first innovation of RDS is its survey design, the affordances of which mitigate some of the challenges of studying hard-to-reach populations. RDS employs a chain-referral design by beginning with a convenience sample of "seed" respondents in the target population; each participant is given a small, fixed number of coupons to distribute to peers also within the target population, who in turn recruit further participants, circumventing the absence of a sampling frame by replacing it with the social network of the population itself. Rather than searching for rare individuals through a broad population frame, RDS enables researchers to leverage the existing social ties of population members who would be prohibitively difficult to reach otherwise. Moreover, RDS' coupon mechanism removes the need for researchers to directly identify or contact members of a stigmatized population, reducing the confidentiality and trust risks that typically depress participation by operationalizing the social trust that exists within the target population.

1.2. Inference

The second innovation of RDS is in conceptualizing the waves of recruitment as a Markov chain where where each participant recruits the next participant from their network at random. It follows from this assumption that sufficient waves of recruitment eventually reduce bias from the convenience sample of seeds; the chain eventually loses memory of where it started such that the composition of the sample is governed by the stationary distribution rather than by the choice of seeds. This strong assumption allows researchers to reason that, adjusting for degree within the target population network, an RDS sample that underwent sufficient waves of recruitment is unbiased.

1.3. Existing Estimators

RDS thus allows researchers to estimate population-level characteristics, where inclusion probabilities can approximated by $\tilde{\pi}_i \propto d_i$, and population-level characteristics can then be estimated by reweighting observations by $1/d_i$.

The standard Volz-Heckathorn (VH) [5], or RDS-II estimator, superseding the Salganik-Heckathorn (SH), or RDS-I, estimator [3], models the recruitment process as a random walk and estimates the population mean of a function $g(Y)$:

$$\hat{\mu}_{VH} = \frac{\sum_{t=1}^{n} \frac{g(Y_{i_t})}{d_{i_t}}}{\sum_{t=1}^{n} \frac{1}{d_{i_t}}}.$$

The VH estimator rests on several key assumptions:

i) The recruitment chain must be long enough for the Markov chain to mix, i.e., the chain must run for more steps than the mixing time of the underlying network, so that the final distribution is approximately stationary and independent of the initial seeds, though this is hardly met in-practice.
ii) Respondents must recruit uniformly at random from their network contacts. In practice, homophily causes respondents to recruit disproportionately from like respondents, so that the recruitment chain in certain settings behaves more like a walk confined to community subgraphs, with rare cross-community transitions.
iii) Self-reported degree must accurately reflect true network degree, since $\hat{y}$ weights each observation by $1/d(i)$, so under-reporting inflates weights while over-reporting deflates them, introducing bias in the direction of misreporting with the outcome y.
iv) The underlying network must be connected and undirected.
v) The random walk modeling lastly assumes sampling with replacement, though in practice RDS is often done without replacement, with practitioners reasoning that this should have minimal impact if the sampling fraction is very small.

Failure to meet any of these assumptions contributes to bias into the final estimates. Assumptions (i) and (ii) contribute severely to bias in the presence of community bottlenecks, as homophilic recruitment keeps the walk within communities, and low conductance means that even a nominally long chain fails to cross community boundaries with sufficient frequency to wash out seed dependence. This motivates the block transition probability framework of Zhang et al. [8] and their post-stratified (PS) estimator, which models inter-community recruitment rates and forms the basis for the bias-correction method we develop here.

When the graph exhibits low conductance – communities are connected by relatively few cross-community edges – the mixing time of the random walk grows large, so short recruitment chains remain concentrated near their starting seeds. This problem may be particularly pronounced in populations with natural community divisions, such as “concentration” in student population, where social ties are dense within communities and sparse across them, exacerbating the “echo-chamber” effect where participants with similar covariates share the same opinion and send the survey to friends with similar covariates.

We are interested in investigating specifically how assumption (ii), uneven recruitment under homophily, holds up in practice, and how it might be addressed if unmet. In this paper, we ask: **How are the assumptions of RDS met and violated in practice, and how well do existing and novel methods for adjusting bias account for these real-world results?** Indeed, how might we de-bias our sample in situations of non-mixing? In the process of accounting for homophily within communities, we extend the Volz-Heckathorn estimator to develop a novel re-weighting mechanism.

2. OUR EXPERIMENT

Artificial Intelligence (AI) has been a massively transformative and fast-moving technology since the initial release of ChatGPT in 2022. A recent Crimson study found that on average, "students say they use artificial intelligence to complete 34.5 percent of their homework" [9]. Given this rapid rollout and adoption of AI, the question of whether or not this technology should exist or be embraced at all are overshadowed by the preeminence of technological accelerationism. Thus, we believe it is interesting and important to give voice to those who oppose AI in order to illuminate alternative viewpoints. Throughout the following, we define *anti-AI* as a self-defined identity against *not anti-AI*, requiring in-network participants to "opt-in" to the label.

2.1. Applicability of RDS

We believe this situation provides an appropriate setting for RDS. There are several challenges to studying anti-AI Harvard students as a target population in practice. For one, anti-AI students are a small population – In the Crimson’s AI study, they found only around 6% of their 300 students surveyed do not use generative AI services [9]. Thus, we can extrapolate that the proportion of Harvard students who are anti-AI is relatively small, and therefore collecting data on these students from a random sample would be quite difficult. Moreover, the identity of anti-AI is stigmatized, especially in particular academic contexts where AI has been adopted as a new norm and reflected in pejoratives like "Luddite." We also assume this population will be practically hard to reach because they’re more likely to be broadly anti-technology and less trusting of institutions and data collection. Lastly, we assume that anti-AI Harvard students exist within a relatively well-connected social network, as college students tend to form friendships within political, cultural, and social groups which likely converge on important societal questions. All of these assumptions make anti-AI Harvard students an appropriate target population to study with RDS: Though they are difficult to study via random sampling, due to small population size and identity stigmatization, their social networks should allow us to utilize referral sampling in order to reach a sufficient number of students to collect and estimate on meaningful data.

Thus, in order to explore our research question about RDS in practice, we utilized RDS to sample anti-AI Harvard undergraduates and collect survey data on **why** they are anti-AI. After collecting data, we will attempt to apply our debiasing estimator to provide a less-biased sample of population opinion, mitigating bias stemming from homophily.

If this were a social science paper, our primary research question would focus on our outcome variable: The rationale for why these students are Anti-AI. However, while we hope to provide some preliminary insights to answering this question through the building of our model, our focused in this statistical research paper is understanding which methodological assumptions can be met in practical applications of RDS and how we might compensate for the violation of our assumptions, particularly that of (i) mixing and (ii) uniform recruitment.

3. SURVEY DESIGN AND DATA PROCESSING

To begin our RDS study, we selected 13 seed nodes and sent them our anti-AI survey. Each respondent was invited to send the survey to up to three referrals. We attempted to mitigate seed bias by choosing seeds across houses, concentrations, and class years. The survey itself consists of five sections: eligibility, anti-AI rationale, demographics, degree estimation, and referral. The full survey can be found in Appendix 8, but here we will go into depth on the relevant **methodological choices** made in designing each aspect of the survey, as well as the **limitations** inherent to the survey design.

3.1. Eligibility

All participants were asked if they identified as anti-AI; an affirmative response was required to proceed through the rest of the survey. 5 of 46 participants (10.9%) did not identify as anti-AI. Their information was excluded from our data analysis as they fell outside the population of interest.

3.2. Anti-AI Rationale

We asked anti-AI participants why they were anti-AI, having them select one of four choices that best reflected their opinion in creating an outcome variable of interest:

1. I have concerns about about its impact on the environment.
2. I don't think the technology is trustworthy and secure.
3. I have concerns about its impact on jobs and economic inequality.
4. I have concerns about its impact on interpersonal relationships.

3.3. Demographics

To identify possible covariates, we asked each participant for their name, political affiliation, and whether they were international. Using the Harvard Facebook, we also collected information on participants' gender, concentration, year, and house. These co-variates were then cleaned and grouped. Houses were grouped into the roughly equal four house neighborhoods of Quad, River West, River Central, and River East (Dudley was included in Quad). We grouped concentrations into the three categories of Arts & Humanities, Social Sciences, and Sciences (or undeclared, as a fourth) – following a taxonomy defined by Harvard [10] – and students were assigned to a category based on their primary concentration (listed first). We believe that these groups will help maintain possible correlates to anti-AI Rationale, as students' overall orientation towards sciences, humanities, or social science informs their worldview and perspectives on new technology. These groupings also maintain roughly balanced sizes, which is especially important given our studies' limited sample size.

3.4. Degree Estimation

We also collected data about the user's number of known connections within the anti-AI network. This is important in order to correct for degree, as RDS is biased towards well-connected nodes who are more likely to be referred. Because the accuracy of self-reported degree can be a source of error in RDS, we made sure to follow best practices to formulate a question that lends itself to accurate self-reporting. As is explained in [7], it is important to ask a specific, measurable proxy question to obtain a relative measure of in-network degree. We arrived at the question: "To the best of your knowledge, what is the number of known anti-AI Harvard students you've communicated with in the last week?". This prevents respondents from lazily overestimating their degree as "communicated with in the last week" enforces a lower bound on the substance of the relation. We also guard against inflation of degree by making sure the potential referees are "known" to be anti-AI. We assume that the risks to underestimate one's degree are low so that these two safeguards contribute only to bias reduction.

3.5. Referrals

Lastly, each participant was asked to send the survey to up to three other participants and provide the full name of those referred. In constructing our referral question, we attempted to lower the bias arising from preferential referrals within networks by asking respondents to "Please **randomly** choose up to three anti-AI Harvard under-

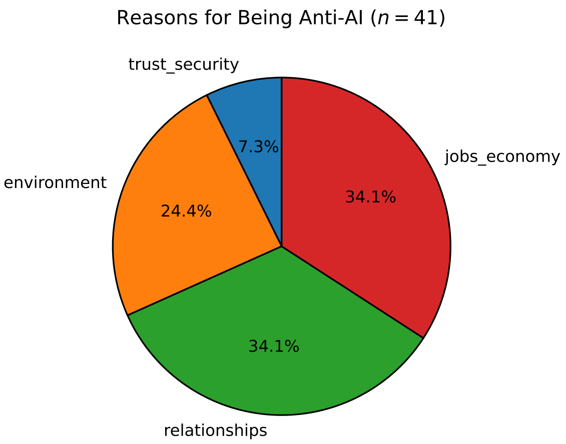

Figure 1. Distribution of reasons for being anti-AI.

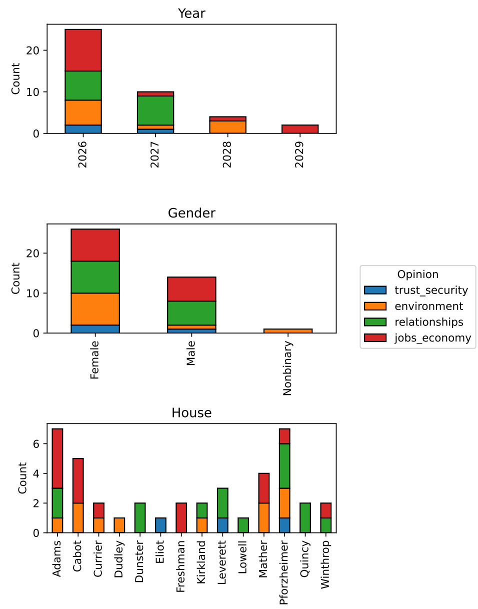

Figure 2. Distribution of respondents across year, gender, and house.

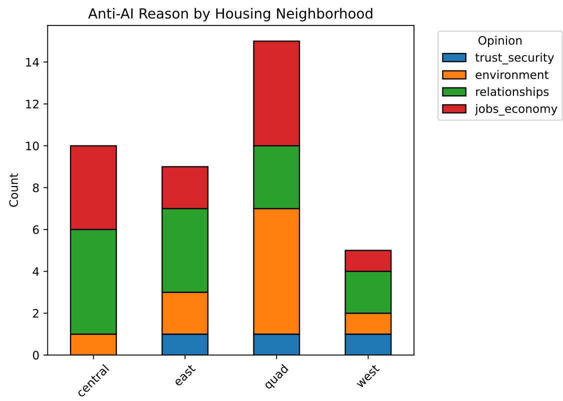

Figure 3. Undergraduate houses bucketed into neighborhoods. Freshmen excluded.

graduates you know to refer for this survey." Nevertheless, we recognize that preferential recruitment likely still occurred.

Our survey incentive, which consisted of compensating the participant \$3 for each of their referrals who successfully completed the survey also likely resulted in bias from preferential referrals, as respondents would be incentivized to send the survey to people who were more likely to fill out the survey, i.e., with whom they were closer. We decided, nevertheless, that this was necessary to maximize the number of survey responses, which was a consistent challenge throughout the study.

4. SURVEY RESULTS

4.1. Summary of Data

We collected 41 samples of anti-AI Harvard students, although we attempted to collect more with nudging emails and reminders, and were limited by non-referral and non-participation. In spite of the relatively small number of participants, which is a risk of dealing with sparse, hard-to-reach populations, we believe the data serves as an interesting example of RDS in practice. First, we see in Fig 1. that anti-AI sentiment is primarily for economic, interpersonal, and environmental reasons, with trust and security being a minor reason.

Next, we see distributions across covariates like year, gender, and house in Fig 2. Upperclassmen are more represented in the sample, which is likely due to seed selection bias since our convenience sample included more upperclassmen, and propagation through the referral process due to preferential recruitment to peers in the same class year. Additionally, the sample contains a higher proportion of women relative to other gender categories. In contrast, the distribution of responses across houses is relatively even, with all houses being represented. These houses are sorted into pre-given neighborhoods in Fig 3; the Quad is slightly overrepresented while River West is slightly underrepresented.

Finally, we see a distribution of respondents over concentration grouped by field in Fig 4. The resulting distribution is relatively even across the three major groups, which is helpful because it will ensure no single academic division disproportionately drives the composition of the sample, and it also means that some amount of mixing has already occurred in the RDS survey run.

4.2. RDS Network

We see the resulting network generated by RDS, clustered and colored by wave at which each individual was recruited in Fig 5. Clearly, the sampling process does not

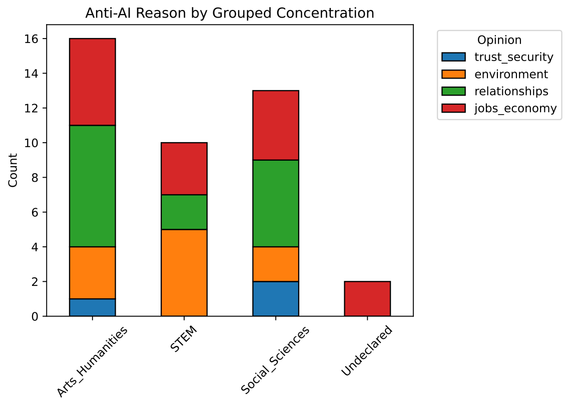

Figure 4. Concentration grouped by field.

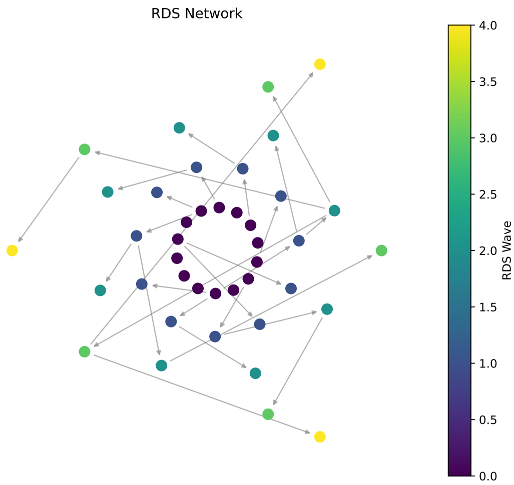

Figure 5. Visualization of the realized RDS network.

form a fully expanding referral tree as we hoped. Many participants (including some seeds) did not refer any additional individuals, and many recruited individuals did not respond to our survey, meaning many recruitment chains terminated early. As a result, the realized sample likely has uneven coverage of the underlying population.

5. METHODOLOGY

5.1. Intuition

We propose a community-aware weighting scheme that generalizes the Volz-Heckathorn estimator by downweighting consecutive same-community referrals, which is motivated by homophily in RDS chains causing referrals to have similar responses, especially when the referred respondent is in the same community as the referrer.

Under strong homophily, the RDS process no longer behaves like a random walk that mixes between communities. In settings where the outcome of interest is strongly correlated with community, consecutive samples within the same community carry less information about the overall population distribution, and cross-community referrals act as mixing events that carry disproportionately more information about population level distributions.

Our idea is to explicitly use this structure in a weighting scheme to reduce the contribution of same-community transitions. Intuitively, this will reduce the effect of oversampling an "echo chamber" where a group of friends with similar covariates share the same opinion and all respond to the survey and then send the survey to more of their friends with similar covariates. While the standard Volz-Heckathorn estimator accounts for bias caused by degree with the assumption of approximate mixing, it doesn't account for the community structure preventing mixing. Similarly, Zhang et al. [8] re-weight based on stratification into predefined groups, but still makes assumptions about mixing. Furthermore, even if strata are well balanced, within-community referral chains could lead to clusters of similar observations. This method is meant to improve estimation by reducing the contribution of redundant observations that could inflate variance.

5.2. Proposed Estimator

Formally, we consider a population represented by a network $G = (V, E)$, partitioned into K communities $\mathcal{C} = \{1, 2, \dots, K\}$. Each individual $i \in V$ has both a community label $c(i) \in \mathcal{C}$ as well as exactly one of I opinions, $Y_i \in \{1, 2, \dots I\}$, which is the ourcome variable of interest. Let the RDS process be defined as a sequence of referrals $\{i_t\}_{t=1}^n$, so the total sample size is n. We propose to downweight within-community referrals; that is,

$$w_t(\alpha) = \frac{1}{d_{i_t}} \cdot \alpha^{I(c(i_t)=c(i_{t-1}))} \tag{1}$$

where $\alpha \in [0, 1]$. Note that seeds as well as first entries into a community are not downweighted, but all following referrals that remain within the same community are downweighted. Then the modified Volz-Heckathorn estimator is

$$\hat{\mu}(\alpha) = \frac{\sum_{t=1}^n w_t(\alpha) g(Y_{i_t})}{\sum_{t=1}^n w_t(\alpha)}.$$

Note that letting $\alpha = 1$ recovers the standard Volz-Heckathorn estimator.

The optimal α should then be determined by community dependence; that is, the correlation of opinions within the same community and the homophily in the network. With high correlation and high homophily, referrals within the same community are likelier to lead to redundant information, meaning that α should be smaller. When correlation and homophily are low, then within-community samples still remain informative, so $\alpha \approx 1$.

We now formally derive an estimate for α. Suppose that for each individual i, we represent the individual's outcome as being composed of global, community, and individual perspectives:

$$g(Y_i) = \mu + \gamma_{c(i)} + \epsilon_i$$

with μ representing a population mean, $\gamma_{c(i)}$ capturing the community-specific deviation, and ϵ_i a user-level noise term. Assume $E[\gamma_{c(i)}] = 0$ and $E[\epsilon_i] = 0$, and the variance decomposes as $\mathrm{Var}(g(Y_i)) = \sigma_\gamma^2 + \sigma_\epsilon^2$. Define the intra-community correlation as $\rho := \frac{\sigma_\gamma^2}{\sigma_\gamma^2+\sigma_\epsilon^2}$. This represents the proportion of variance explained by community-level opinions. Then for two nodes i, j, we have:

$$\mathrm{Cov}(g(Y_i, Y_j)) = \begin{cases} \sigma_\gamma^2, & c(i) = c(j) \\ 0, & \text{otherwise.} \end{cases}$$

Suppose we have k consecutive samples all from the same community. Let $\bar{g}_k = \frac{1}{k}\sum_{t=1}^{k} g(Y_{i_t})$. This has variance

$$\mathrm{Var}(\bar{g}_k) = \frac{1}{k^2}\sum_{t=1}^{k}\sum_{s=1}^{k}\mathrm{Cov}(g(Y_{i_t}, Y_{i_s})).$$

Then using the variance and covariance from above, we have that

$$\begin{aligned}\mathrm{Var}(\bar{g}_k) &= \frac{1}{k^2}[k(\sigma_\gamma^2 + \sigma_\epsilon^2) + k(k-1)\sigma_\gamma^2] \\ &= \frac{\sigma_\gamma^2 + \sigma_\epsilon^2}{k}(1 + (k-1)\rho).\end{aligned}$$

We define the effective sample size n_{eff} as the number of independent samples that would give the same variance as the correlated sample mean $\mathrm{Var}(\bar{g}_k)$; therefore, we solve

$$\begin{aligned}\frac{\sigma_\gamma^2 + \sigma_\epsilon^2}{k}(1 + (k-1)\rho) &= \frac{\sigma_\gamma^2 + \sigma_\epsilon^2}{n_{eff}} \\ \Rightarrow\ n_{eff} &= \frac{k}{1 + (k-1)\rho}.\end{aligned}$$

Ideally, the total contribution of a run of k samples in the same community should have weight similar to its effective sample size n_{eff}. Because each observation is assigned weight α (besides the first observation, but we ignore this), then the total weight of the run is $k\alpha$. Then solving

$$\begin{aligned}k\alpha &\approx \frac{k}{1 + (k-1)\rho} \\ \Rightarrow\ \alpha^* &\approx \frac{1}{1 + (k-1)\rho}.\end{aligned}$$

This value of α^* makes sense in limiting cases: when $\rho = 0$, all samples are important because intra-community referrals are as informative as any referral; this gives $\alpha^* = 1$ as expected. When $k = 1$, we have $\alpha^* = 1$ which again makes sense because the original entry into a community shouldn't be downweighted. Finally, when ρ, k are both large, then $\alpha^* \to 0$ which is logical because with high correlation and high homophily, new referrals within the same community are less informative and should be discounted at a higher rate.

The above derivation is dependent on assumptions like communities being observed, correlation primarily being driven by community (which appears in the formulation for $g(Y_i)$), and the sampling structure being well-approximated by runs of consecutive samples in a single community of typical length k.

6. EMPIRICAL TESTING

6.1. Experimental Setup

We use simulated networks generated from stochastic block models (SBMs) to evaluate the performance of our weighting estimator. We do this because we have control over levels of homophily and correlation of opinion with community, and we can also generate the ground-truth community and opinion labels. This allows us to initially explore the behavior of our estimator outside the limitations of our empirical study.

We first consider an SBM with 7 blocks and 120 total nodes. To introduce realistic heterogeneity in community sizes, we do not make equal block proportions, but assign nodes to communities by sampling a probability vector from a Dirichlet distribution and assigning nodes proportionally to the sampled probability vector. Edges are then generated according to the SBM with intracommunity edge probability $p_{in} = 0.5$ and intercommunity edge probability $p_{out} = 0.02$, giving a strongly associative graph. There are four opinions (encoded 0-3); each community is assigned a designated dominant opinion, sampled uniformly from the four opinions. Each node in the community's opinion is the same as the community dominant opinion with probability 90%, and it is any of the other three opinions with equal probability (3.3%).

Let us consider the value of α^* derived from the above specifications. We chose to have a 90% correlation of individual opinion with community dominant opinion, so we have $\rho = 0.9$. Furthermore, we let k be the expected length of consecutive samples in a single community. We estimate the referral process to be such that each person refers exactly one other person. Then we estimate the probability that the referred person is in the same community by considering a node's expected number of neighbors in the same community divided by that node's total expected neighbors. If node i_t is in community m with N total nodes and n_c nodes in community c, then

$$P(c(i_t) = c(i_{t-1})) \approx \frac{n_m p_{in}}{n_m p_{in} + (N - n_m)p_{out}}.$$

For simplicity's sake, we estimate that $n_m p_{in} + (N - n_m)p_{out} \approx n_k$. This is because we are applying this method in a regime where we have high community cohesion ($p_{in} \to 1$) and low inter-community connection ($p_{out} \to 0$), meaning that the first term dominates, and we have the n_m approximation. Therefore since $P(c(i_t) = c(i_{t-1})) \approx p_{in}$, the length of a consecutive run within the same community follows a first success distribution with parameter $1 - p_{in}$ so

$$k \approx \frac{1}{1 - p_{in}}$$

which gives

$$\alpha^* = \frac{1}{1 + (k-1)\rho} = \frac{1}{1 + \frac{p_{in}}{1-p_{in}}\rho}.$$

Here, because $p_{in} = 0.5$ and $\rho = 0.9$, we have that $\alpha^* \approx 0.526$.

We simulate RDS on each generated network using 2 initial seed nodes; each respondent then recruits 2 neighbors, and sampling continues until a total sample size of 30 nodes is reached. This setup creates a controlled environment with strong correlation of community with opinion as well as high homophily.

Because the entire population's distribution is known, we can use Kullback-Leibler (KL) divergence to measure discrepancy between true population distribution and the distribution recovered by each estimator. Let P denote the true distribution of opinions in the full SBM population, and $\hat{P}$ denote an estimated distribution. Then

$$D_{KL}(P||\hat{P}) = \sum_x P(x) \log\left(\frac{P(x)}{\hat{P}(x)}\right).$$

KL divergence is useful because we're interested in comparing full categorical distributions, and we want to penalize misallocation in any category. Furthermore, the asymmetry in KL divergence is helpful here because if certain groups with less popular opinions are systematically undersampled (making $\hat{P}(x)$ smaller), then KL divergence will place a larger penalty compared to overestimating those same groups. This is especially relevant in the RDS setting where we aim to have a fair representation of the population's true opinion distribution.

6.2. Experimental Results

We first see a single result of the SBM graph generation and estimator correction. The table below compares the KL divergence between each quantity and the true population distribution, and a curve comparing KL divergence to different values of α is shown in Fig 6.

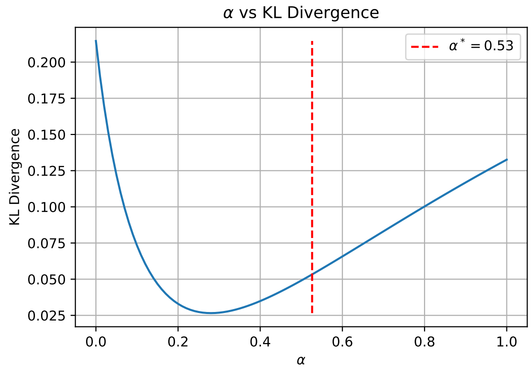

Figure 6. Setting $\alpha = \alpha^*$ reduces KL divergence compared to $\alpha = 1$, the vanilla Volz-Heckathorn estimator.

	KL Divergence
Sample Frequency vs. Population	0.287278
VH vs. Population	0.132597
Adjusted VH vs. Population	0.053308

Unsurprisingly, the raw sampled frequencies show the largest deviation from true population distribution. The Volz-Heckathorn estimate reduces error by accounting for degree-related sampling bias, but still performs worse than the adjusted estimator that we proposed. This single-trial result is consistent with our hypothesis that explicitly accounting for community-related correlation can reduce the distributional bias more than the vanilla Volz-Heckathorn estimator.

We now consider a larger experiment of 100 samples; each sample is generated independently with the same SBM generation and RDS simulation as in the previous part. In each trial, we compare the KL divergence of the estimator at $\alpha = 1$ (no adjustment to Volz-Heckathorn) with the KL divergence at $\alpha = \alpha^*$, with α^* as found by our heuristic in 5.2. Specifically, we compute the difference

$$D_{KL}(P||\hat{P}_{\alpha=1}) - D_{KL}(P||\hat{P}_{\alpha^*})$$

so a positive value means that an improvement has occurred using α^*, and larger values mean greater reduction in distributional error achieved by the proposed method.

The resulting histogram is shown in Fig 7. Across the 100 trials, we see that the proposed method yields a positive improvement in KL divergence in 68% of cases. The mean difference in KL divergence is 0.0198, indicating that on average, modifying α with our heuristic reduces distributional error. Both the moderately high success rate and positive mean improvement are promising, and the adjustment seems capable of reduce the underlying bias in the sampling process with some consistency. Still though, the gains are not uniformly large, and 32% of trials exhibit worsened performance using α^*. While the heuristic may be beneficial often, it is not universally op-

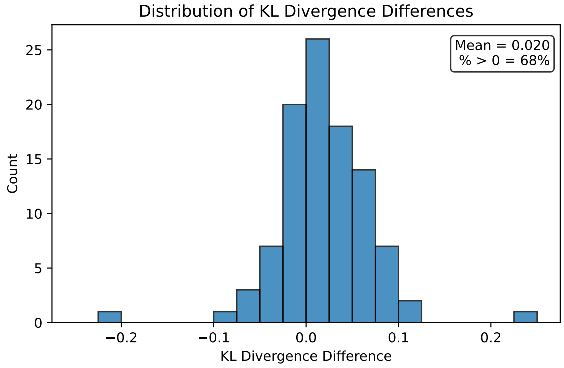

Figure 7. Distribution of KL divergence differences across 100 trials, where positive values indicate improvement from the proposed adjustment.

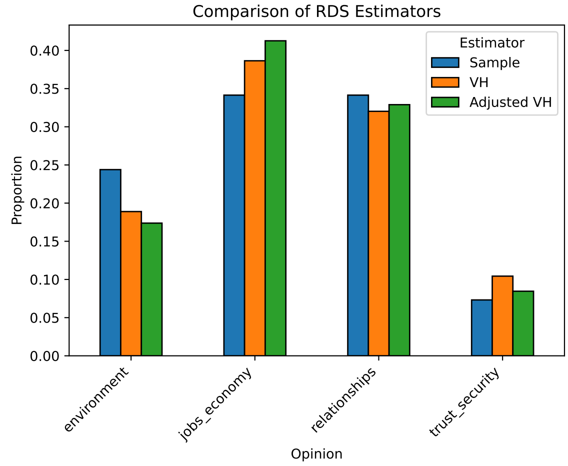

Figure 8. Distribution of opinions based on RDS estimator.

timal and seems quite sensitive to the specific realization of both the network and the sampling process.

7. APPLICATION ON DATA

7.1. Deriving the Estimator

In applying our methodology to our Anti-AI survey data, we identified **grouped concentrations** as the most relevant community to be controlled for. We choose concentration as the community forming attribute because we suspect that individuals are much more likely to refer other students within their concentration group, and because we suspect that concentration is influential in informing the perspective and worldview behind the individual's rationale. This intuition is supported by the data: of the covariates we suspect might correlate with students' rationale, we found that concentration had the highest modularity as shown in the Appendix Table 2.

To derive our estimator, we must find the appropriate reweighting constant α. We will use the formula for α^* which requires that we have estimates for p_{in} and ρ. We will model the network as an SBM with community structure defined by the grouped concentrations (using first listed, or primary concentration). These community assignments can then be used to estimate homophily. We also estimate the correlation of community opinion with the dominant opinion of that community. To estimate p_{in} and p_{out}, we use the observed RDS recruitment edges. Let E_{in} be the number of edges where the recruiter and recruit are in the same community, and E_{out} be the number of edges where recruiter and recruit are in different communities. Then we find

$$\hat{p}_{in} = \frac{E_{in}}{E_{in} + E_{out}}, \hat{p}_{out} = \frac{E_{out}}{E_{in} + E_{out}}.$$

Of course, these probabilities are not literal SBM edge probabilities, but instead an empirical measure of tendency of within-community recruitment. It is, however, impossible for us to recover the true underlying friendship network. Therefore, we'll treat $\hat{p}_{in}$ as the intracommunity connection probability of an SBM. Running this algorithm gives $\hat{p}_{in} = 0.555$.

Next, we estimate ρ, the correlation of an individual's opinion with the dominant opinion of the community. For each community, we first identify the dominant opinion as the most common response of those community members. We then compute the fraction of individuals whose opinion matches the dominant opinion. Formally, if o_i is the opinion of individual i and $o^*(c(i))$ is the dominant opinion of community $c(i)$, then

$$\hat{\rho} = \frac{1}{N} \sum_{i=1}^{N} I(o_i = o^*(c(i))).$$

This algorithm gives $\hat{\rho} = 0.463$, and $\alpha^* = 0.633$.

7.2. Analyzing Estimator Performance

We then run our proposed method on the data with the derived $\alpha^* = 0.633$. The results appear in Fig 8; we can see that the sample distribution, Volz-Heckathorn estimator, and our proposed estimator all give different distributions for anti-AI sentiment. However, the results of Volz-Heckathorn and our proposed method yield very similar results, which is expected because α is moderately large. The close alignment of all three estimates suggests that, in this setting, Volz-Heckathorn is already able to account for the majority of distortion induced by RDS, and the α adjustment doesn't add much improvement. The moderate estimates of $\hat{p}_{in}$ and $\hat{\rho}$ indicate that while there is within-group preference for recruitment as well as general correlation with a "dominant" choice per concentration, it is not sufficiently strong to cause clustering of opinions on recruitment chains. As a result, the sampling process is able to have enough mixing that degree correction alone may be sufficient to correct for bias.

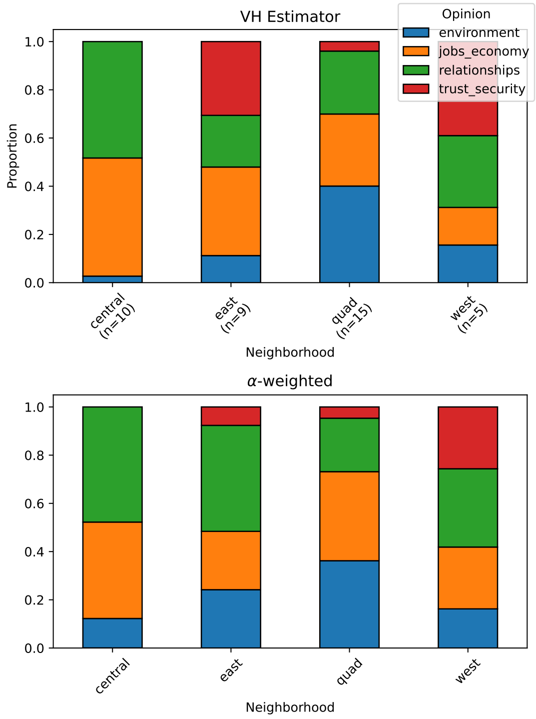

Figure 9. Distribution of opinions by neighborhood when weighted with Volz-Heckathorn vs. our proposed method. Using α-weighting gives more similar distributions between the four neighborhoods. Note small samples represented.

7.3. Robustness Checks

We next perform robustness checks of our method. While we do not know the true distribution of beliefs among the Harvard undergraduate population, we will evaluate our α-weighted correction by cross-neighborhood heterogeneity. Specifically, we expect that anti-AI reasoning should be independent of one's physical living neighborhood, which is largely random, apart from minor exceptions such as transfers). Therefore, systematic differences in observed opinion distribution across neighborhoods are more likely due to errors in sampling than due to actual variation in beliefs.

Under this interpretation, a desirable effect of our estimator is to reduce variation in conditional distributions across neighborhoods. To investigate this, we consider the distribution of opinion in each neighorhood, but we weight opinion using weights from Volz-Heckathorn or our proposed α-weighting method. As we can see in Fig 9, using our methodology makes distribution of opinion more similar across neighborhoods. To quantify this effect, we measure the dispersion of neighborhood-level distributions relative to the overall distribution generated for each estimator method. Specifically, for each neighborhood, we compute the squared ℓ_2 distance between its opinion distribution and the global average distribution, and then average this quantity across neighborhoods; we do this for raw sample, the Volz–Heckathorn estimator, and our α-adjusted estimator.

Estimator	ℓ_2 Dispersion
Raw Sample	0.0364
Volz–Heckathorn (VH)	0.0723
α-weighted	0.0326

Table 1. Cross-neighborhood dispersion of opinion distributions under different estimators. Lower values indicate greater similarity across neighborhoods.

As seen in Table 1, the ℓ_2 dispersion increases when applying the Volz–Heckathorn estimator relative to the raw sample, meaning degree-based reweighting amplifies differences in opinion distributions across neighborhoods. However, our α-weighted estimator gives lowest dispersion of the three, meaning that heterogeneity across the four neighborhoods has been reduced. This gives evidence that RDS does indeed introduce dependence on clustering that cannot be addressed by degree correction alone, and our α adjustment can partially correct for sampling biases by making neighborhood-level distributions more similar.

7.4. Discussion of Results

Across both examined estimators as well as the raw data, it is clear that *trust_security* concerns account for the smallest proportion of the primary reason why anti-AI students at Harvard self-identify as anti-AI. The categories *environment* and *relationships* see greater representation in α-weighting, while the other two categories see lesser representation. Both the VH and α-weighted results suggest that Harvard students are less concerned about abstract fears around safety and misuse (which is somewhat surprising given the growth of clubs like the Harvard AI Safety Student Team), and instead have a greater fear of impact on social structures and long-term societal outcomes.

The increased proportion of *environment* and *relationships* under the α-weighted estimator indicates that these concerns are held more broadly across the network, while the other two categories appear more within concentrated groups. Together, the results suggest that Harvard students fear the consequences of AI proliferating in every aspect of life; interpersonal interactions, labor and inequality, and the environment could all be reshaped by AI. Taking these broader concerns into account should be a priority for those who wish to see more robust student adoption of AI services and deeper integration of AI into learning at Harvard.

Due to limitations of sample size and Markov chain runtime, we proceed cautiously in attempting to extrapolate much further. As mentioned, we struggled with survey participation in non-responsiveness and non-referrals, despite our relatively-large number of seeds. With more time and resources, a future RDS study could increase incentives to create a larger network that exhibits a greater amount of successive waves. Although this led to the termination of certain chains, we also noticed that participants referred people who did not identify as anti-AI five times, which may suggest that some who are perceived to be anti-AI do not actually self-identify as anti-AI. Future study of this population may turn to qualitative analysis and further disaggregation of anti-AI students' opinions, recognizing that our outcomes may just reflect survey options and do not leave space for more nuanced beliefs.

8. FUTURE DIRECTIONS

Our proposed α-weighted correction demonstrates promise in aligning RDS distributions more closely to true population-level distributions and reducing spurious heterogeneity induced by the RDS sampling process. In particular, our robustness check suggests that the method can mitigate cross-neighborhood variation due to recruitment dependence in applied settings. However, as observed in our simulation studies, the magnitude of these improvements is often modest. In many graphs, especially those that have moderate homophily and weak correlation between outcome and network structure, the Volz-Heckathorn estimator already reduces the majority of sampling bias, leaving limited scope for additional corrections. The practical gains from adjustments are inherently data-dependent and are small in well-mixed populations.

A key direction for future work is to relax several strong assumptions underlying our framework. First, the derivation of α^* relies on very many simplified structural assumptions about the sampling process, including approximations to referral dynamics and the use of summary parameters such as $\hat{\rho}$ and $\hat{p}_{in}$. These quantities are themselves estimated from data and may be sensitive to model specification and sampling variability, and the estimations of these parameters could also be made more precise (especially the estimation of $\hat{p}_{in}$ from RDS); future study could attempt to independently compute $\hat{\rho}$ to derive α^*. Additionally, our formulation implicitly assumes a relatively homogeneous referral mechanism, such as individuals referring a fixed or similar number of peers, which does not hold in practice as observed in our data where some individuals made no referrals.

Furthermore, our model treats individuals as belonging to a single latent community, whereas an individual may be part of multiple communities (e.g. by being a double concentrator). Using only groups of academic concentration may be too strong and unrealistic of a simplification, because many other factors like social network and background presumably factor into one's opinions. Extending the framework to account for overlapping or mixed community membership would allow for a more realistic representation of the underlying network and may improve the accuracy of dependence corrections.

More broadly, these results indicate that while correcting for sampling bias in certain regimes is theoretically important, its empirical impact depends substantially on network structure and the outcome's relationship to the network structure. Future work could focus on identifying regimes in which such corrections are most beneficial, as well as developing adaptive methods that adjust the strength of correction based on observed network properties.

For our study of anti-AI students, we'd also like to study how other covariates such as political affiliation relate to anti-AI sentiment. However, this was difficult to do in our data. Unsurprisingly, the samples collected leaned significantly towards Democrats (over 60%) with no Republicans, making it impossible for us to draw any meaningful comparisons across political parties. This imbalance also highlights that our α-weighting scheme can only partially correct for bias when there is at least some representation from multiple groups in the sample. In the extreme case where sampling is only from a single group, there is no cross-group information to exploit, and our method can overcome the lack of representation in data.

Our methodology provides a step toward incorporating explicitly covariate dependence-aware corrections into RDS estimation, but further work is needed to relax simplifying assumptions, better estimate underlying structural parameters, and understand when such corrections make meaningful improvements.

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APPENDIX: ANTI-AI SURVEY INSTRUMENT

Survey Overview

This survey was administered via Google Forms. Respondents with anti-AI views were recruited through a snowball sampling procedure: each participant was asked to refer up to three anti-AI Harvard undergraduates, and was compensated $3 per successful referral (limited to the first 70 respondents). The survey link was https://forms.gle/4QndvJVYefKf7YwG8.

Note: Contact emails for the research team: `akang@college.harvard.edu`, `jjpoulson@college.harvard.edu`, `justinli@college.harvard.edu`, `bobbycupps@college.harvard.edu`, `zgladney@college.harvard.edu`.

Survey Questions

Screening Question

1. Do you consider yourself anti-AI? *(Required)*
   - Yes ☐
   - No ☐

   Respondents who answered "No" did not proceed further.

Main Survey *(Shown only to respondents who answered "Yes" above)*

2. Why are you anti-AI? Please choose the response that most closely reflects your opinion. *(Required)*
   - I have concerns about its impact on the environment. ☐
   - I don't think the technology is trustworthy and secure. ☐
   - I have concerns about its impact on jobs and economic inequality. ☐
   - I have concerns about its impact on interpersonal relationships. ☐
3. What political party do you most closely identify with? *(Required)*
   - Democratic ☐
   - Republican ☐
   - Independent ☐
   - Other ☐
   - Prefer not to say ☐

4. Do you consider yourself international? *(Required)*
    - Yes ☐
    - No ☐
5. To the best of your knowledge, what is the number of known anti-AI Harvard students you have communicated with in the last week? *(Required, numeric response)*
6. What is your Zelle or Venmo (please specify which)? *(Optional)*
   Compensation was provided for successful referrals only. If a referred respondent was referred by multiple people, only the first referrer was compensated.
7. Please list the names of up to three anti-AI Harvard undergraduates you are referring to this survey. *(Required: Name #1; Optional: Names #2–3)*
8. Additional comments? *(Optional)*

   Modularity Table *(Using covariates to define communities)*

Table 2. Q_{ER} uses an Erdős–Rényi null model; Q_{Newman} uses the configuration-model null.

Covariate	# Groups	Q_{ER}	Q_{Newman}
Concentration (grouped)	4	0.241	0.230
Political party	4	0.121	0.144
International	2	0.028	0.030
Gender	3	−0.075	−0.054